

WOW!

Although vaccines are available now for many diseases, there is still no vaccine for the common cold.

THE JET INJECTOR

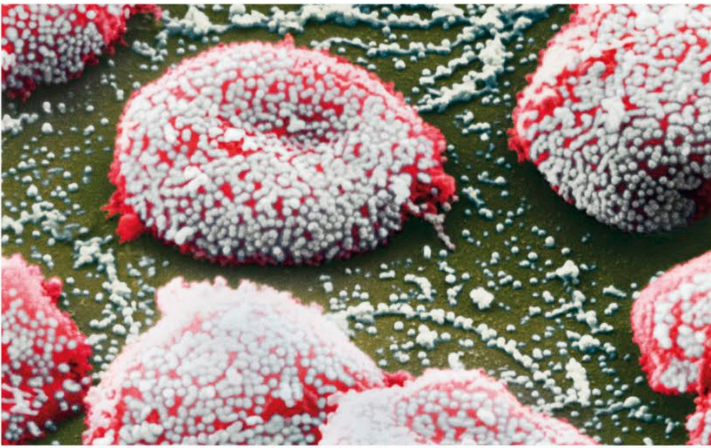
Invented in 1936 by the American engineer Marshall Lockhart, the jet injector delivers medicine as a thin, high-pressure jet of liquid that can pierce skin. It can be operated very quickly, which has led to it being used in mass vaccination programs. But because of the risk of infection, it has been replaced in recent times by disposable, single-use versions.



Flu vaccine being injected using a jet injector

IN GOOD HEALTH

Flu virus particles (shown in white, below) attack red blood cells



DEFEATING DISEASES

The search for new vaccines is ongoing. In the 1940s, the first flu vaccines appeared, followed by vaccines against hepatitis A in the 1990s, and malaria in 2018. However, there are still many deadly diseases, such as Ebola and AIDS, that cannot be vaccinated against.

MODERN VACCINES

In the 20th century, more vaccines were developed, including those against tuberculosis (1921), measles (1963), and rubella (1966). In the 1950s, the American physician Jonas Salk created a polio vaccine but refused to patent it so as to make it freely available to everyone.

▼ ORAL VACCINATIONS

The oral polio vaccine, being given here to a girl in Yemen was invented by the American physician Albert Sabin in the 1960s.



FAST FACTS

- Once one of the world's deadliest diseases, smallpox killed around 400,000 people a year in Europe in the late 18th century.
- In 1980, the World Health Organization (WHO) declared that smallpox had been successfully wiped out by vaccination; so far, it is the only infectious disease to have been eradicated.